# Relatório do projeto de software

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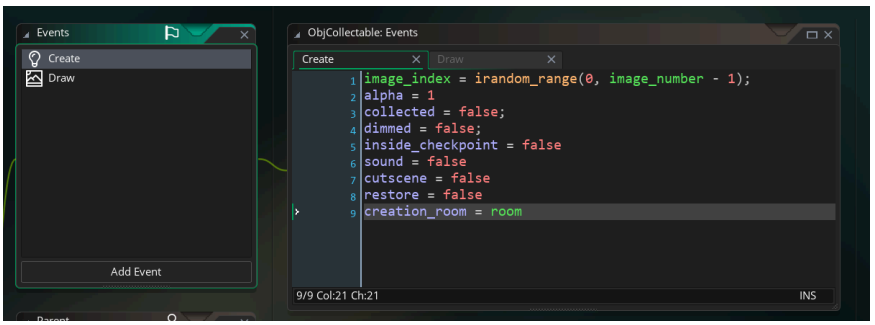
Projeto: The World's Hardest Game: Remix
Linguagem do projeto: GML
Ambiente do projeto: GameMaker

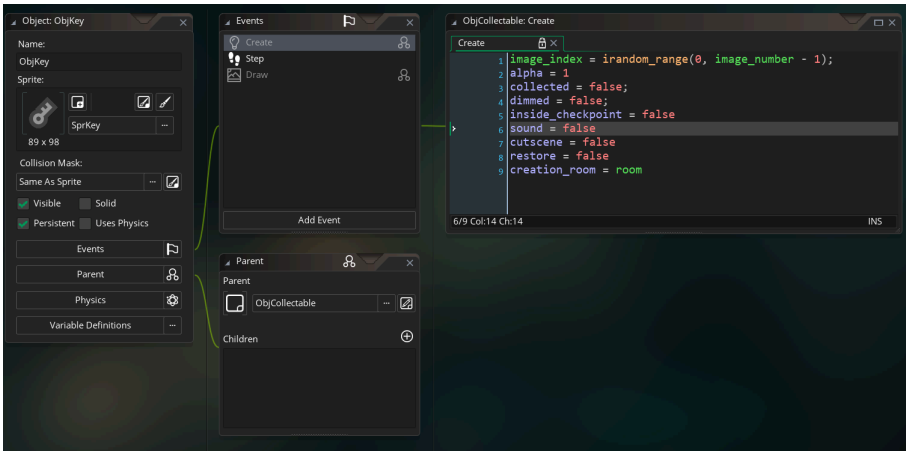
## Herança

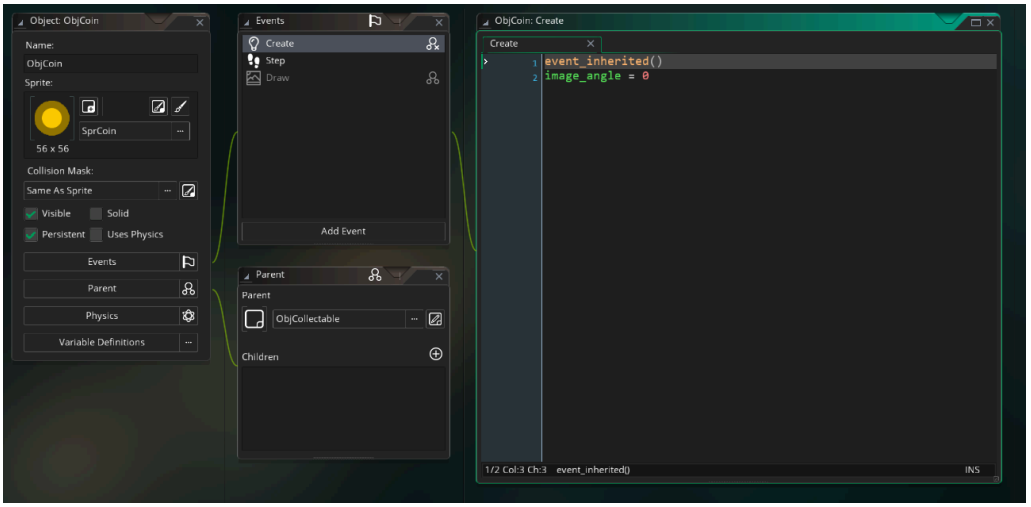
No ambiente GameMaker, a herança ocorre com objetos cujo pai é outro objeto preexistente, herdando os mesmos eventos do objeto pai. Caso se queira sobrescrever o evento herdado mantendo a herança, deve-se chamar a função event_inherited.

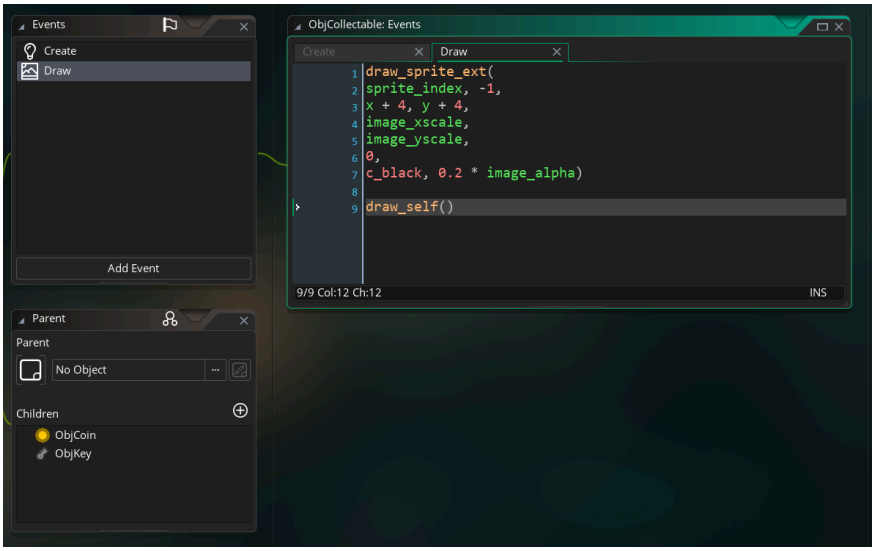
**Ocorrências de herança**

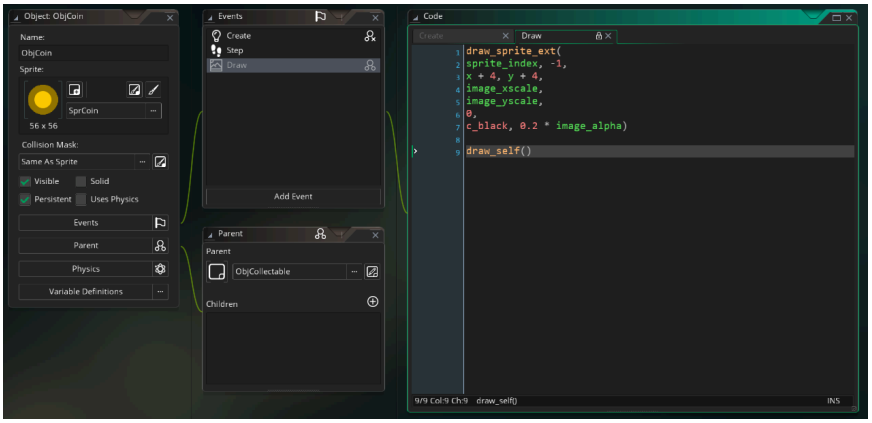
1. ObjCollectable (filhos: ObjKey, ObjCoin)

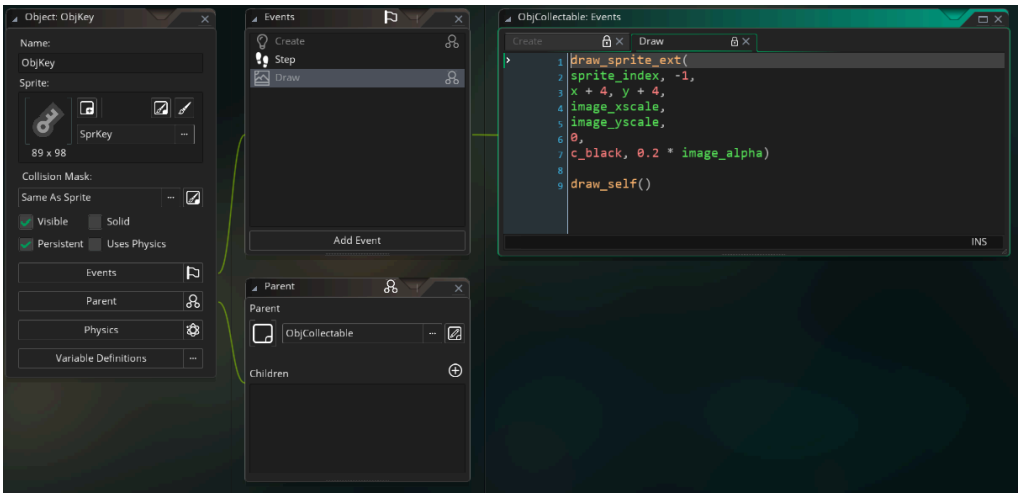


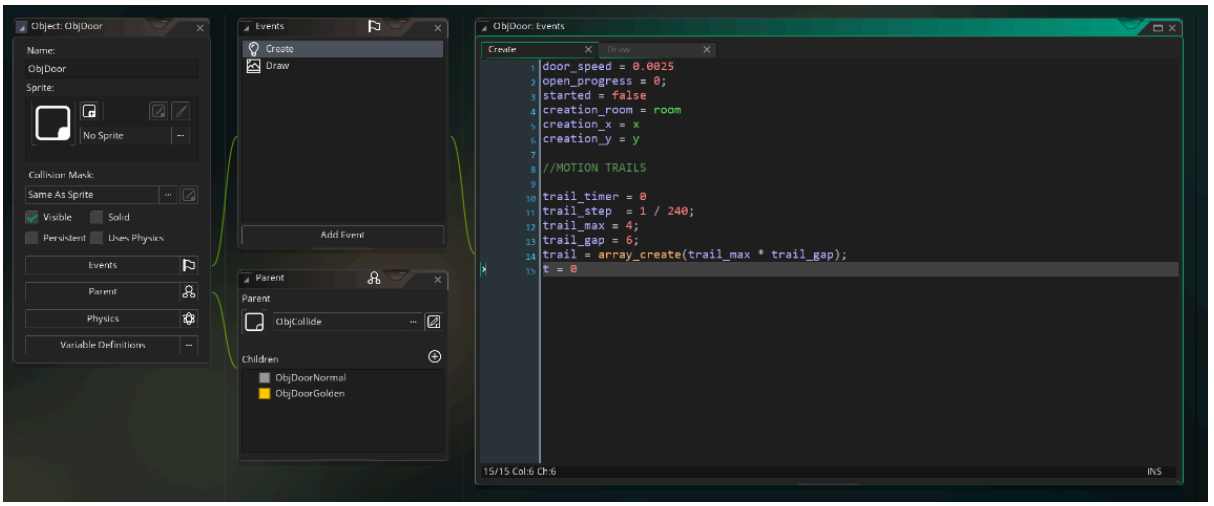



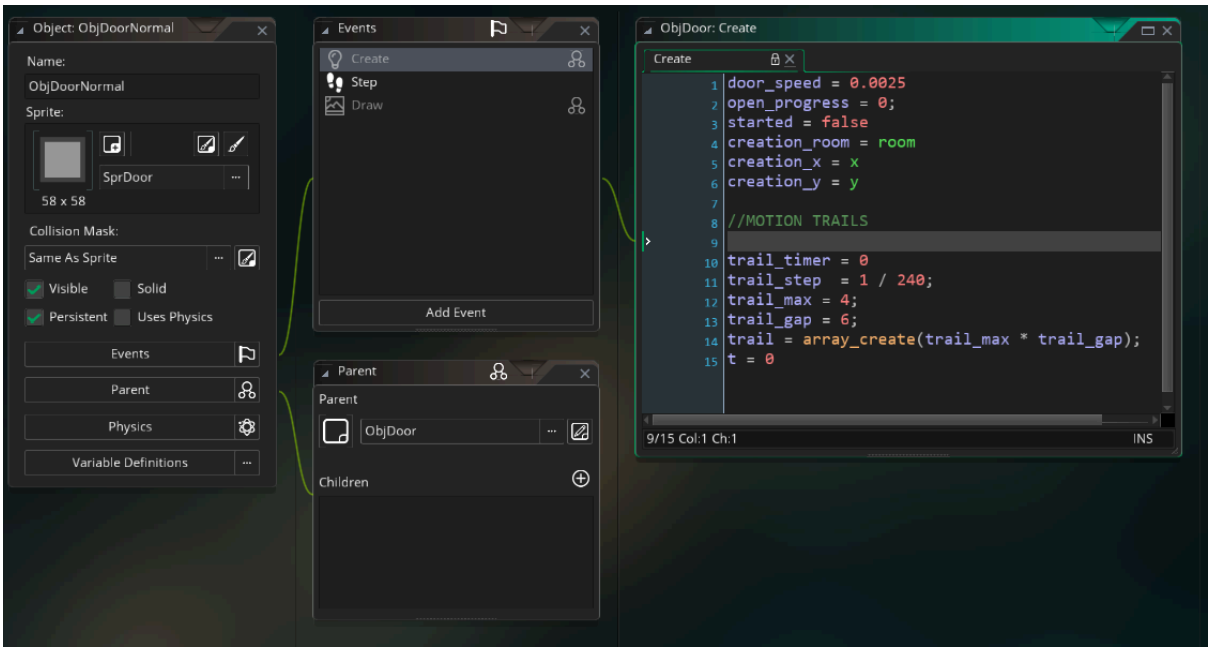

Events
Create
Draw
Add Event
Parent
Parent
No Object
Children
ObjCoin
ObjKey
ObjCollectable: Events
Create
Draw
draw_sprite_ext(
sprite_index, -1,
x + 4, y + 4,
image_xscale,
image_yscale,
0,
c_black, 0.2 * image_alpha)
draw_self()
9/9 Col:12 Ch:12
INS

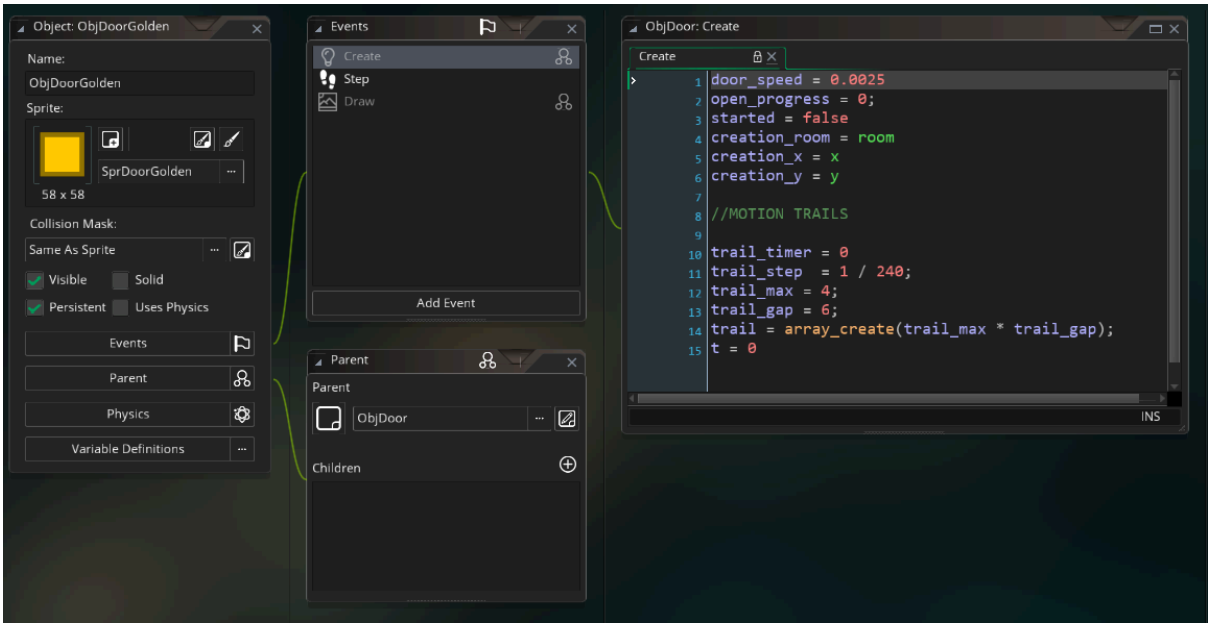

Object: ObjCoin
Name:
ObjCoin
Sprite:
SprCoin
56 x 56
Collision Mask:
Same As Sprite
Visible
Solid
Persistent
Uses Physics
Events
Parent
Physics
Variable Definitions
Events
Create
Step
Draw
Add Event
Parent
Parent
ObjCollectable
Children
Code
Create
Draw
draw_sprite_ext(
sprite_index, -1,
x + 4, y + 4,
image_xscale,
image_yscale,
0,
c_black, 0.2 * image_alpha)
draw_self()
9/9 Col:9 Ch:9 draw_self()
INS

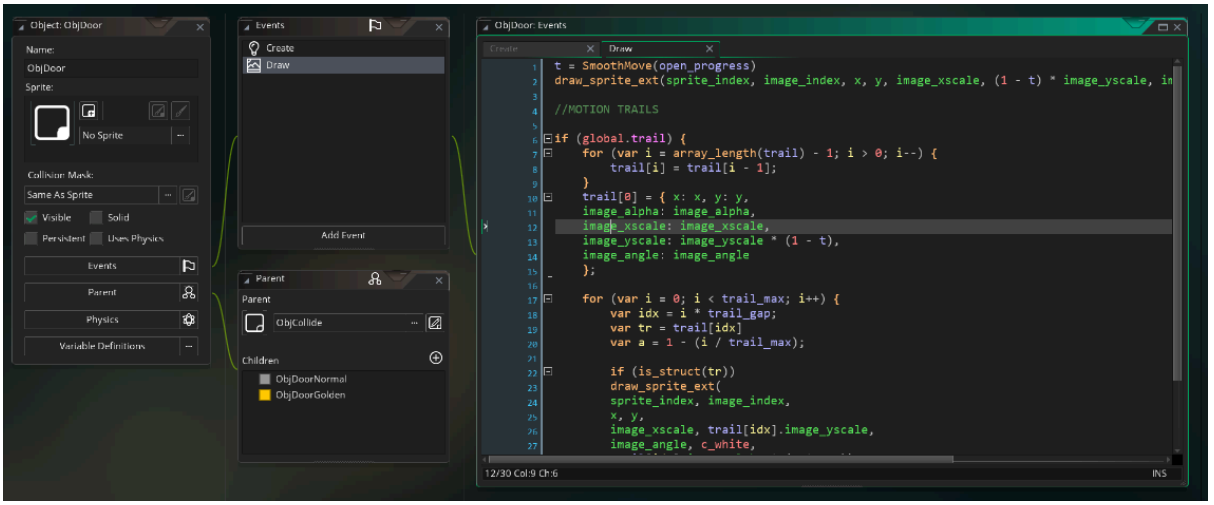

Object: ObjKey
Name:
ObjKey
Sprite:
SprKey
89 x 98
Collision Mask:
Same As Sprite
Visible
Solid
Persistent
Uses Physics
Events
Parent
Physics
Variable Definitions
Events
Create
Step
Draw
Add Event
Parent
Parent
ObjCollectable
Children
ObjCollectable: Events
Create
Draw
draw_sprite_ext(
sprite_index, -1,
x + 4, y + 4,
image_xscale,
image_yscale,
0,
c_black, 0.2 * image_alpha)
draw_self()
INS

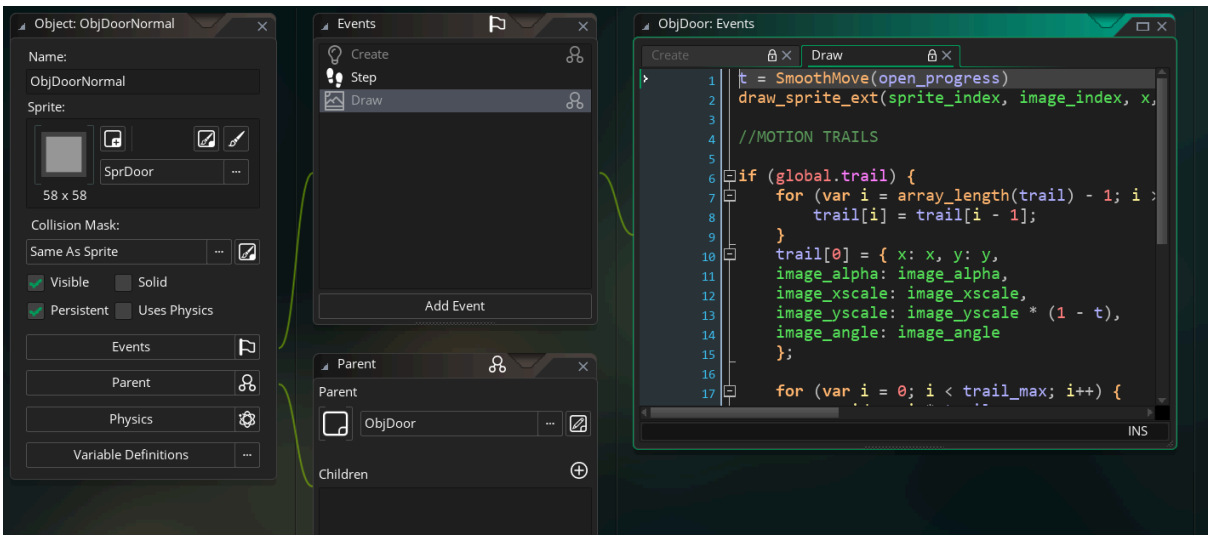
2. ObjDoor (filhos: ObjDoorNormal, ObjDoorGolden)

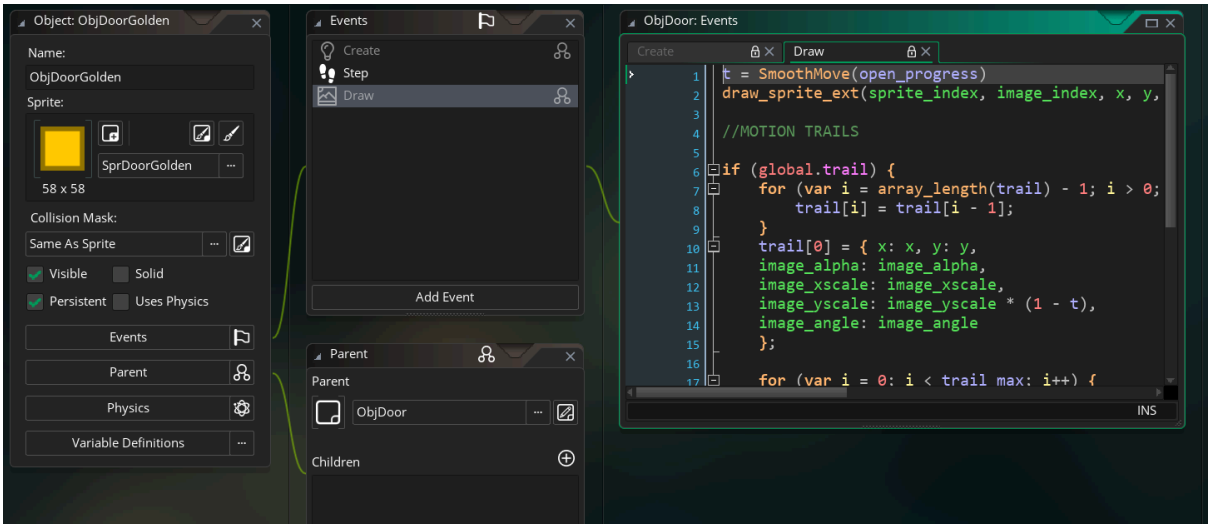


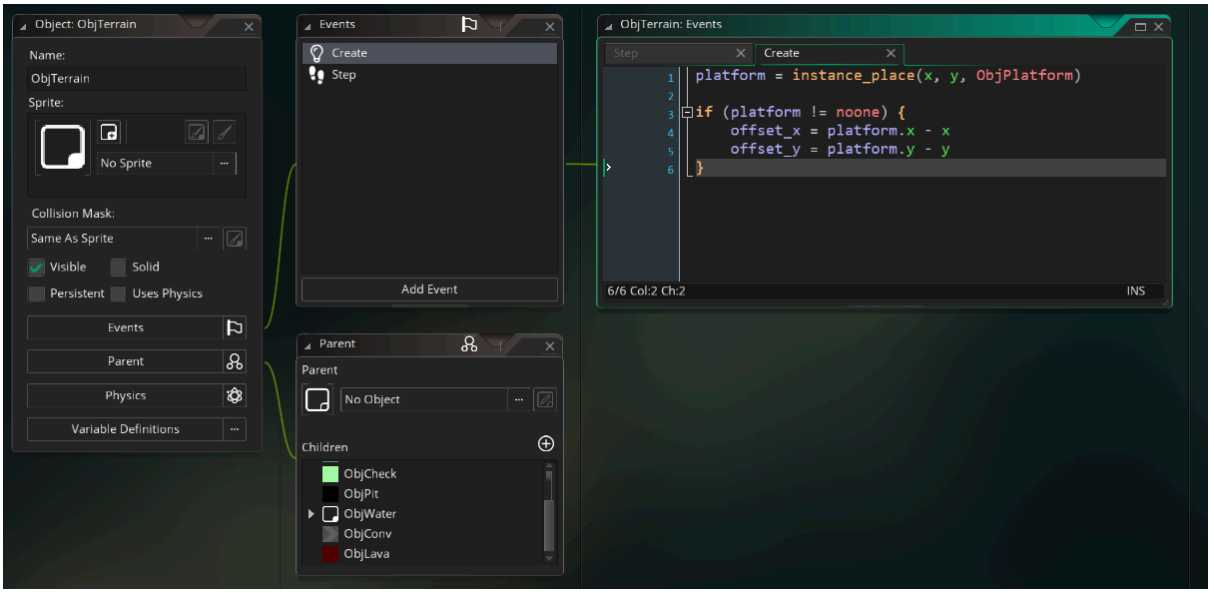



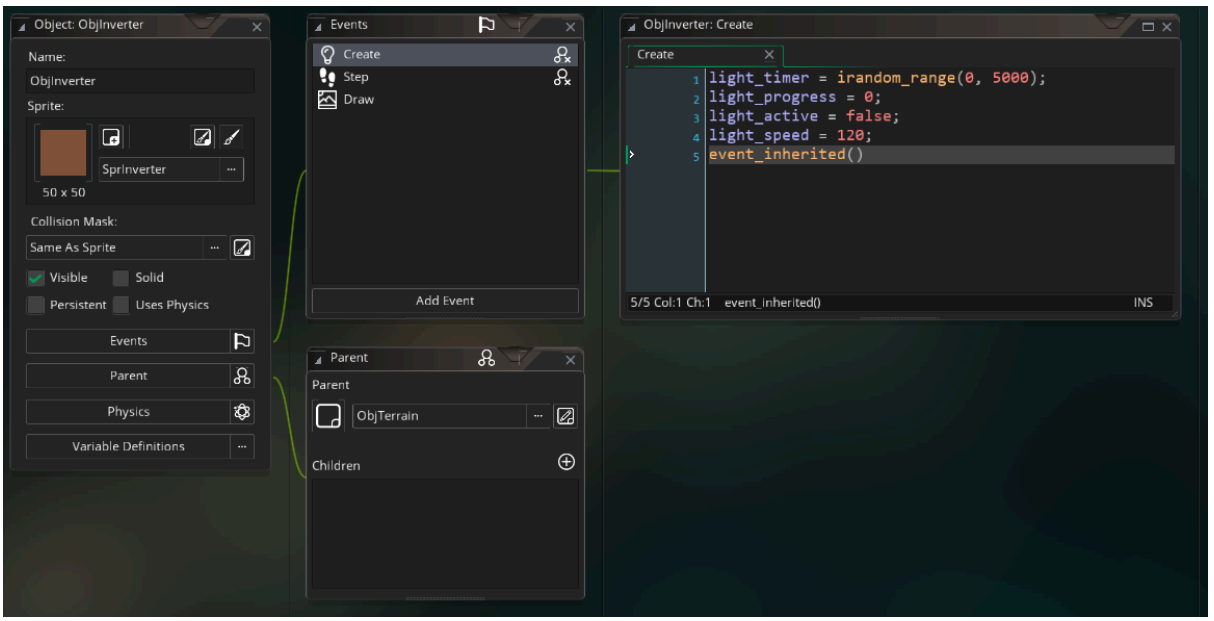

Object: ObjDoor
Name:
ObjDoor
Sprite:
No Sprite
Collision Mask:
Same As Sprite
Visible
Solid
Persistent
Uses Physics
Events
Parent
Physics
Variable Definitions
Events
Create
Draw
Add Event
Parent
Parent
ObjCollide
Children
ObjDoorNormal
ObjDoorGolden
ObjDoor: Events
Create
Draw
t = SmoothMove(open_progress)
draw_sprite_ext(sprite_index, image_index, x, y, image_xscale, (1 - t) * image_yscale, im
//MOTION TRAILS
if (global.trail) {
for (var i = array_length(trail) - 1; i > 0; i--) {
trail[i] = trail[i - 1];
}
trail[0] = { x: x, y: y,
image_alpha: image_alpha,
image_xscale: image_xscale,
image_yscale: image_yscale * (1 - t),
image_angle: image_angle
};
for (var i = 0; i < trail_max; i++) {
var idx = i * trail_gap;
var tr = trail[idx]
var a = 1 - (i / trail_max);
if (is_struct(tr))
draw_sprite_ext(
sprite_index, image_index,
x, y,
image_xscale, trail[idx].image_yscale,
image_angle, c_white,
12/30 Col:9 Ch:6
INS

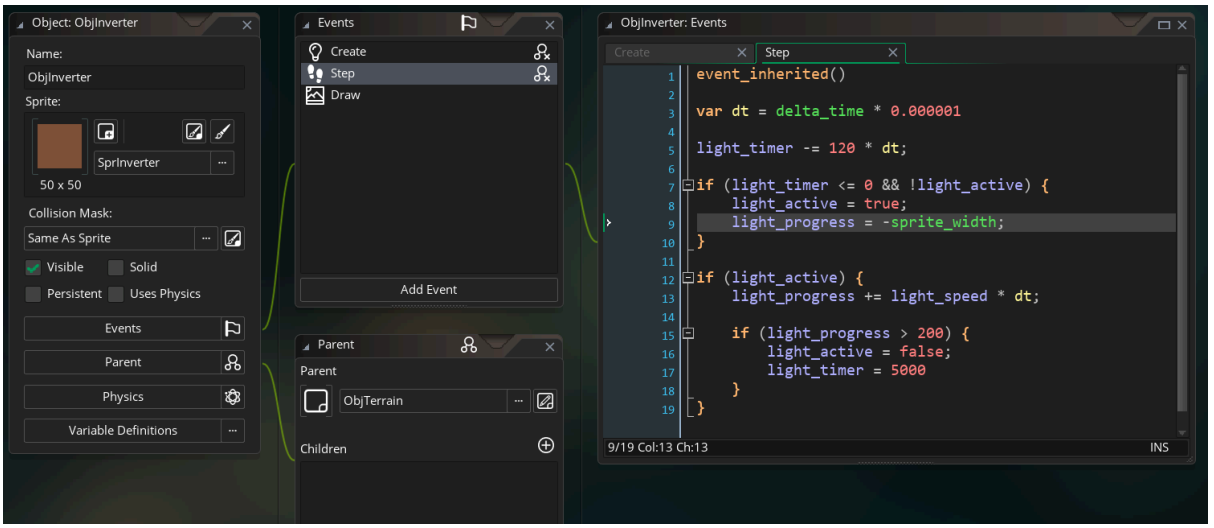

Object: ObjDoorNormal
Name:
ObjDoorNormal
Sprite:
SprDoor
58 x 58
Collision Mask:
Same As Sprite
Visible
Solid
Persistent
Uses Physics
Events
Parent
Physics
Variable Definitions
Events
Create
Step
Draw
Add Event
Parent
Parent
ObjDoor
Children
ObjDoor: Events
Create
Draw
t = SmoothMove(open_progress)
draw_sprite_ext(sprite_index, image_index, x,
//MOTION TRAILS
if (global.trail) {
for (var i = array_length(trail) - 1; i >
trail[i] = trail[i - 1];
}
trail[0] = { x: x, y: y,
image_alpha: image_alpha,
image_xscale: image_xscale,
image_yscale: image_yscale * (1 - t),
image_angle: image_angle
};
for (var i = 0; i < trail_max; i++) {
INS

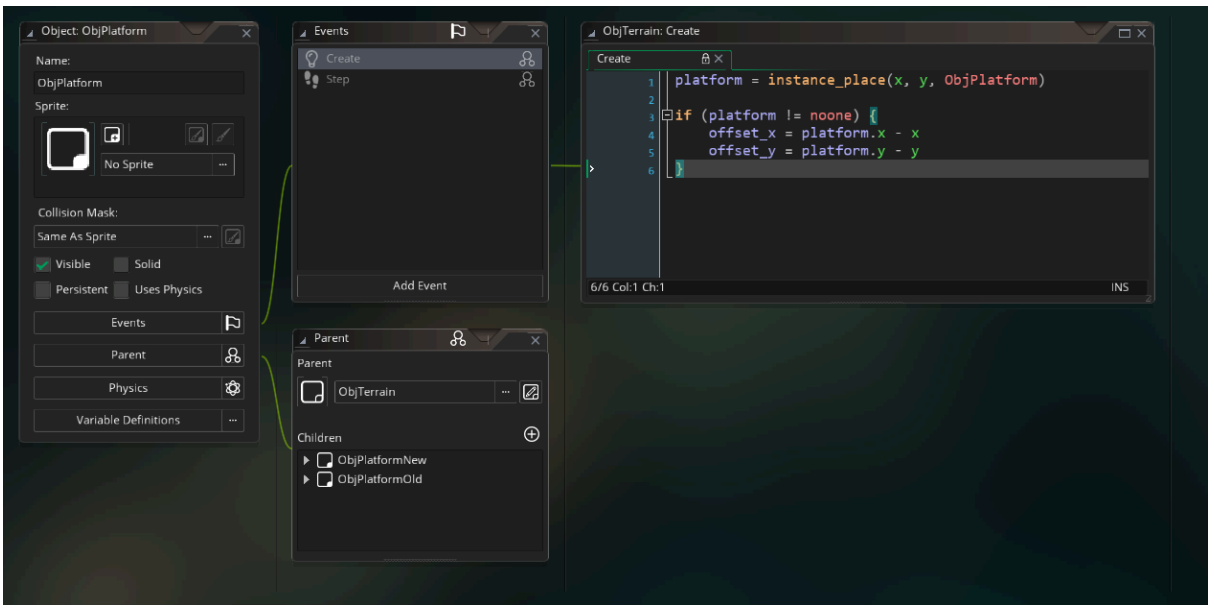

Object: ObjDoorGolden
Name:
ObjDoorGolden
Sprite:
SprDoorGolden
58 x 58
Collision Mask:
Same As Sprite
Visible
Solid
Persistent
Uses Physics
Events
Parent
Physics
Variable Definitions
Events
Create
Step
Draw
Add Event
Parent
Parent
ObjDoor
Children
ObjDoor: Events
Create
Draw
t = SmoothMove(open_progress)
draw_sprite_ext(sprite_index, image_index, x, y,
//MOTION TRAILS
if (global.trail) {
for (var i = array_length(trail) - 1; i > 0;
trail[i] = trail[i - 1];
}
trail[0] = { x: x, y: y,
image_alpha: image_alpha,
image_xscale: image_xscale,
image_yscale: image_yscale * (1 - t),
image_angle: image_angle
};
for (var i = 0; i < trail_max; i++) {
INS

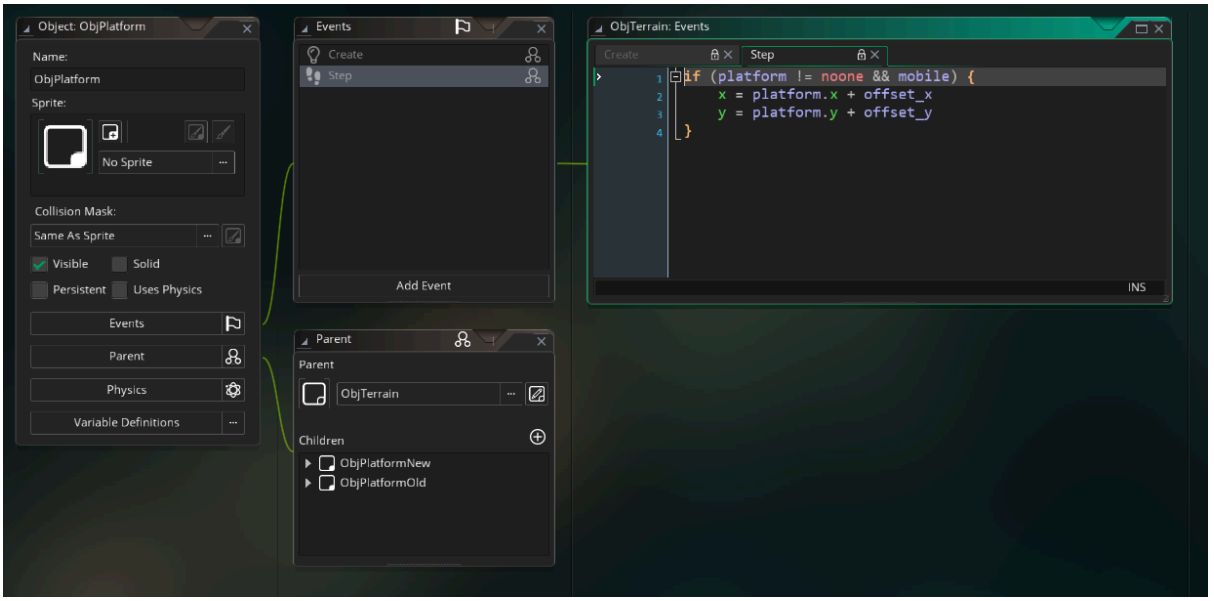
3. ObjTerrain (filhos: ObjInverter, ObjPlatform, ObjIce, ObjCheck, ObjPit, ObjWater, ObjConv, ObjLava)

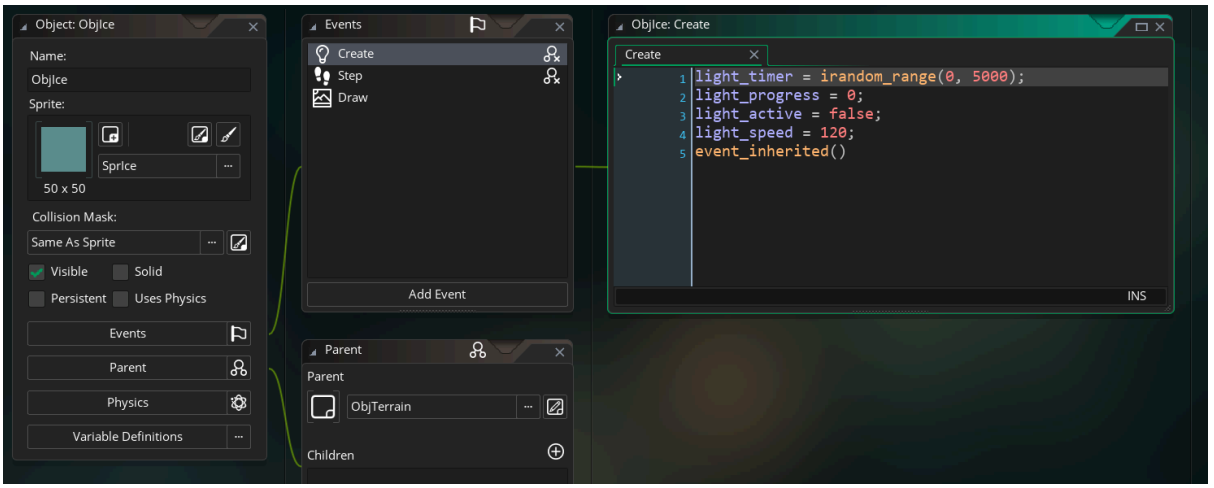


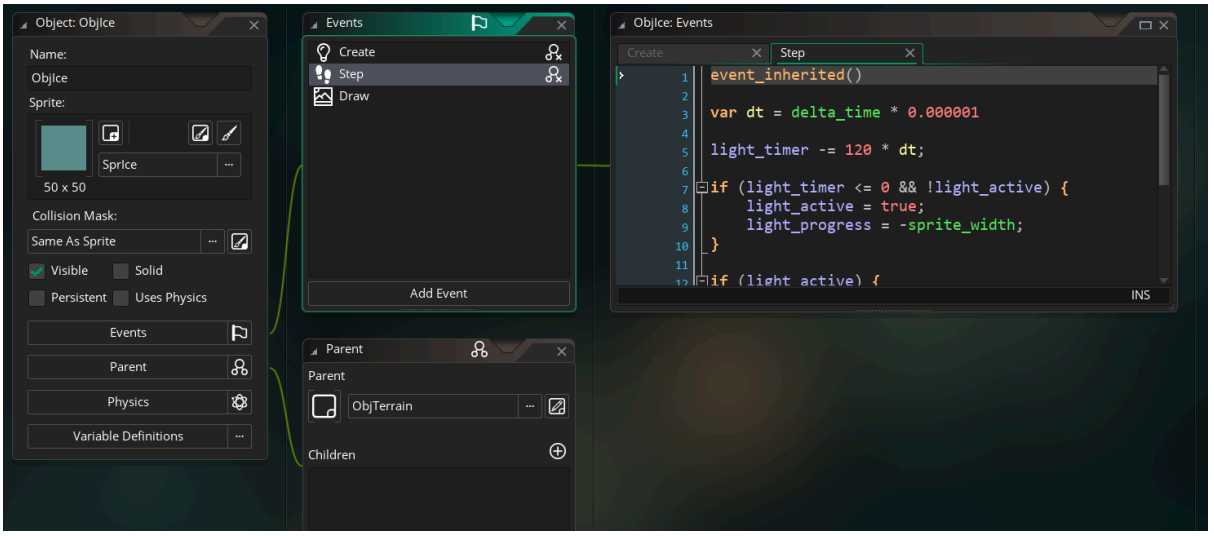



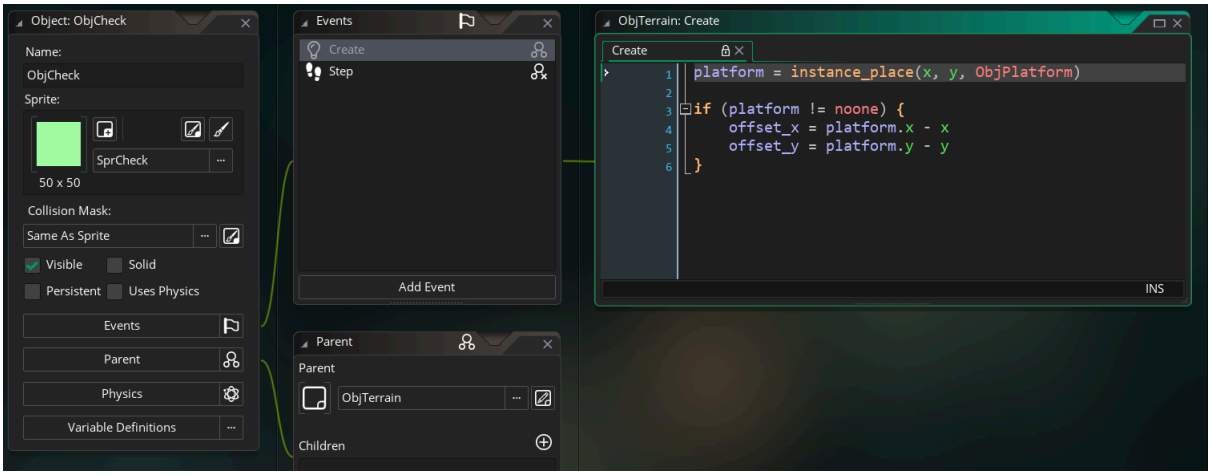



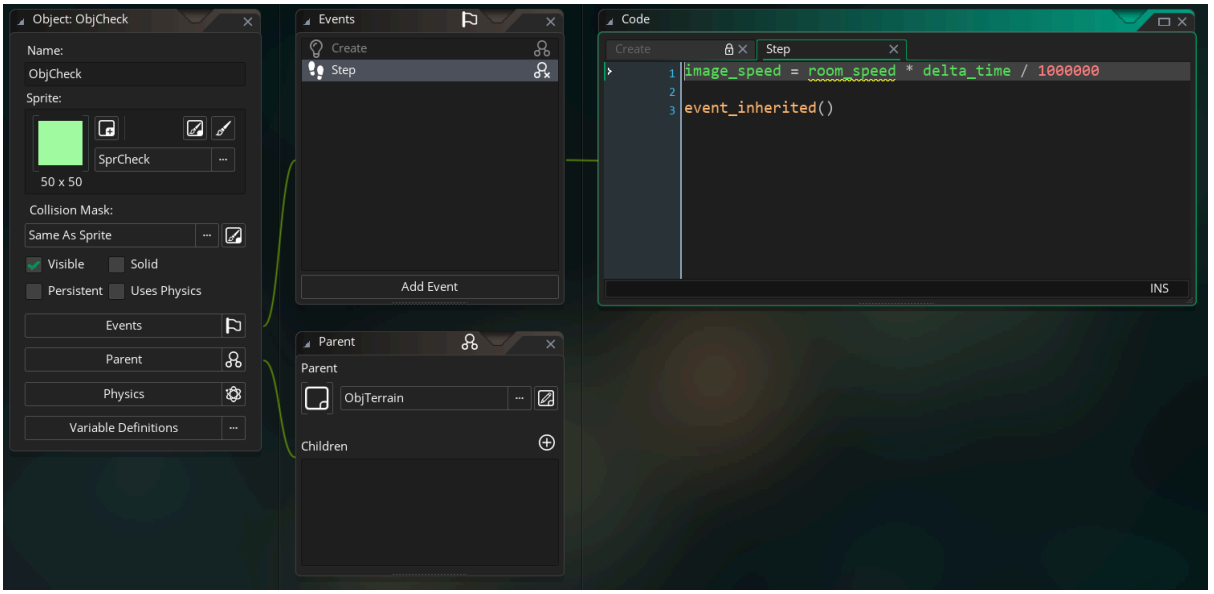

Object: ObjPlatform
Name:
ObjPlatform
Sprite:
No Sprite
Collision Mask:
Same As Sprite
Visible
Solid
Persistent
Uses Physics
Events
Parent
Physics
Variable Definitions
Events
Create
Step
Add Event
Parent
Parent
ObjTerrain
Children
ObjPlatformNew
ObjPlatformOld
ObjTerrain: Create
Create
platform = instance_place(x, y, ObjPlatform)
if (platform != noone) {
offset_x = platform.x - x
offset_y = platform.y - y
}
6/6 Col:1 Ch:1
INS

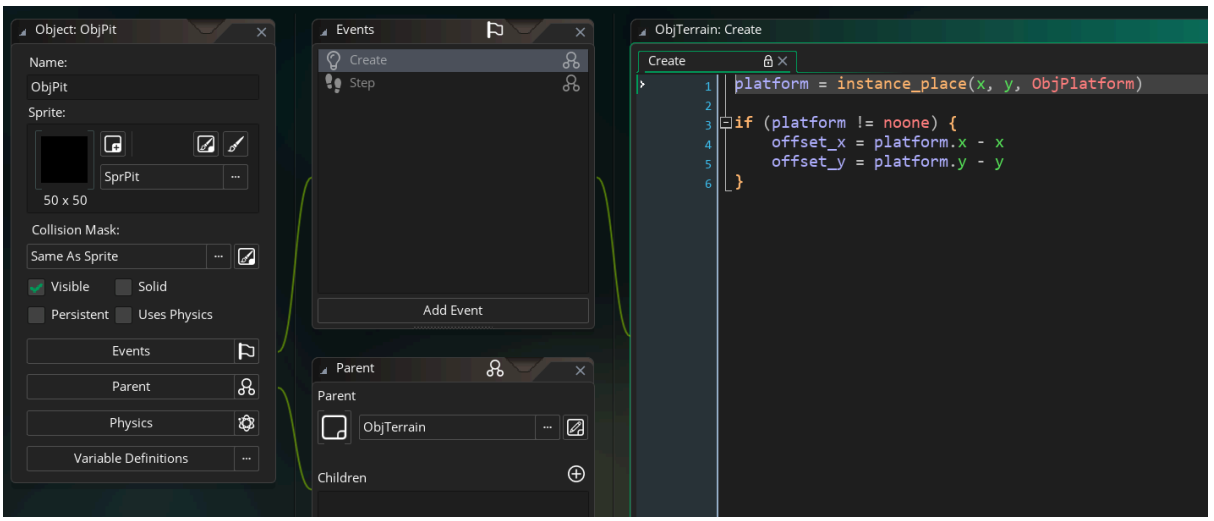

Object: ObjPlatform
Name:
ObjPlatform
Sprite:
No Sprite
Collision Mask:
Same As Sprite
Visible
Solid
Persistent
Uses Physics
Events
Parent
Physics
Variable Definitions
Events
Create
Step
Add Event
Parent
Parent
ObjTerrain
Children
ObjPlatformNew
ObjPlatformOld
ObjTerrain: Events
Create
Step
if (platform != noone && mobile) {
x = platform.x + offset_x
y = platform.y + offset_y
}
INS

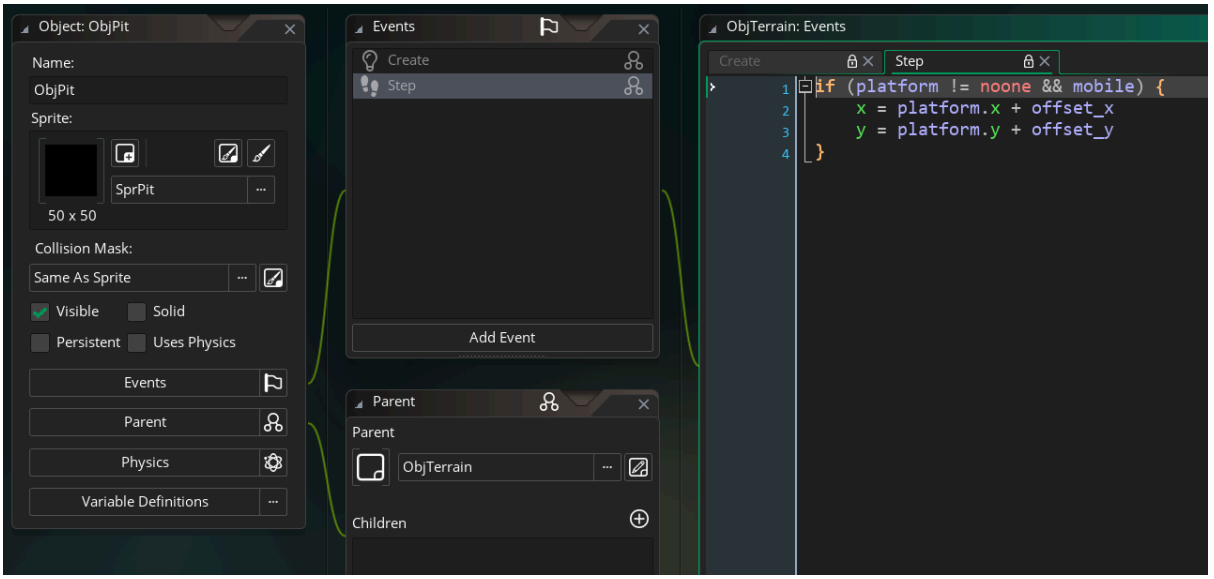

Object: ObjIce
Name:
ObjIce
Sprite:
SprIce
50 x 50
Collision Mask:
Same As Sprite
Visible
Solid
Persistent
Uses Physics
Events
Parent
Physics
Variable Definitions
Events
Create
Step
Draw
Add Event
Parent
Parent
ObjTerrain
Children
ObjIce: Create
Create
light_timer = irandom_range(0, 5000);
light_progress = 0;
light_active = false;
light_speed = 120;
event_inherited()
INS

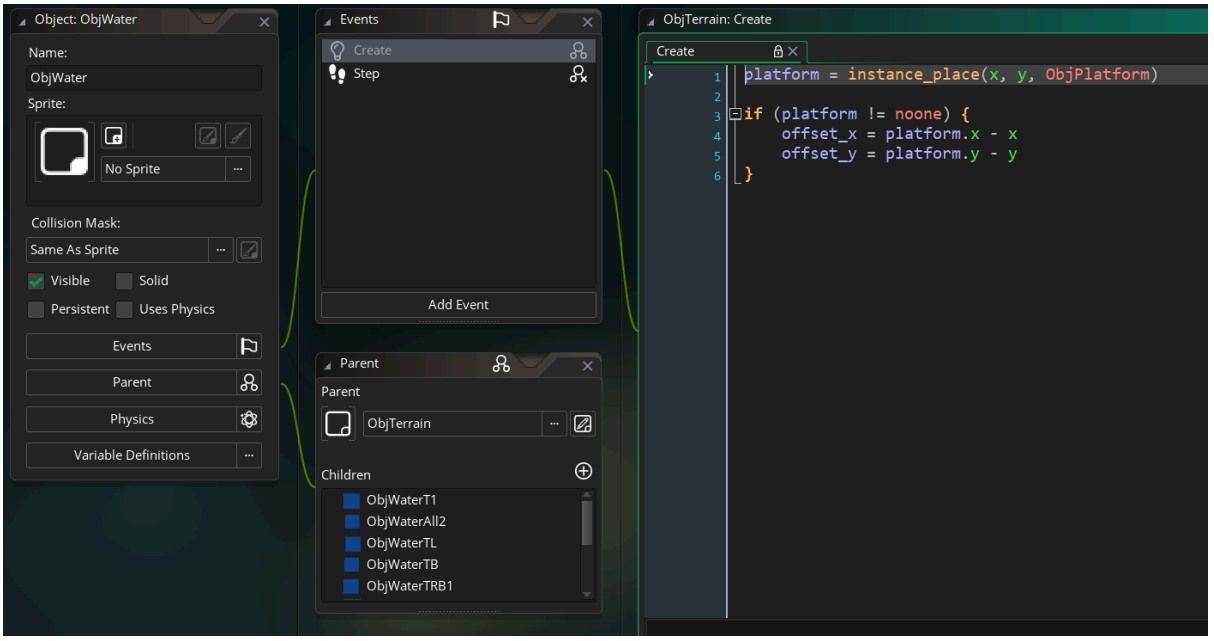

Object: ObjIce
Name:
ObjIce
Sprite:
SprIce
50 x 50
Collision Mask:
Same As Sprite
Visible
Solid
Persistent
Uses Physics
Events
Parent
Physics
Variable Definitions
Events
Create
Step
Draw
Add Event
Parent
Parent
ObjTerrain
Children
ObjIce: Events
Create
Step
event_inherited()
var dt = delta_time * 0.000001
light_timer -= 120 * dt;
if (light_timer <= 0 && !light_active) {
light_active = true;
light_progress = -sprite_width;
}
if (light_active) {
INS

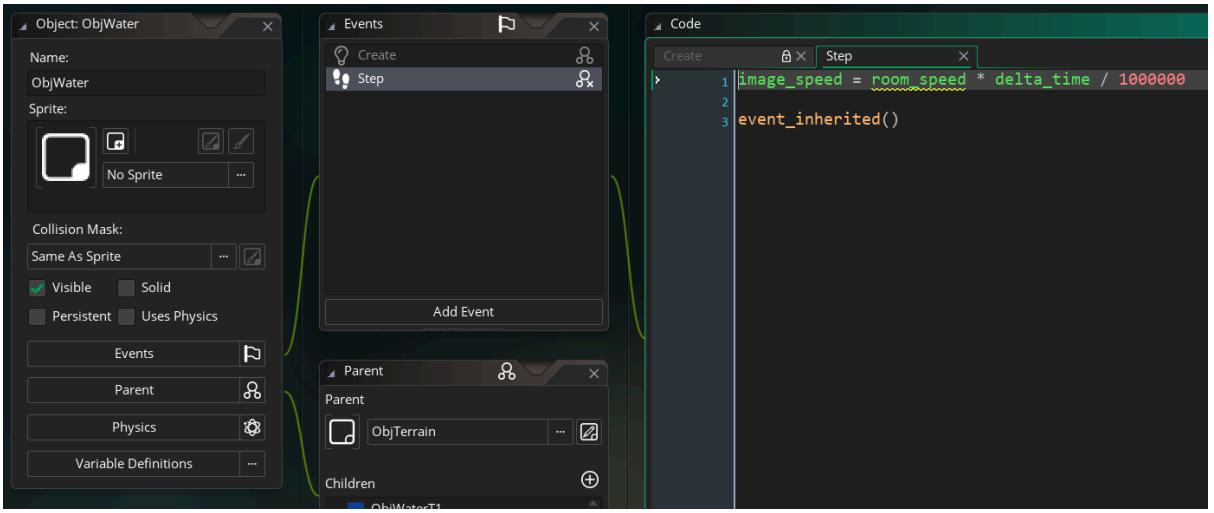

Object: ObjCheck
Name:
ObjCheck
Sprite:
SprCheck
50 x 50
Collision Mask:
Same As Sprite
Visible
Solid
Persistent
Uses Physics
Events
Parent
Physics
Variable Definitions
Events
Create
Step
Add Event
Parent
Parent
ObjTerrain
Children
ObjTerrain: Create
Create
platform = instance_place(x, y, ObjPlatform)
if (platform != noone) {
offset_x = platform.x - x
offset_y = platform.y - y
}
INS

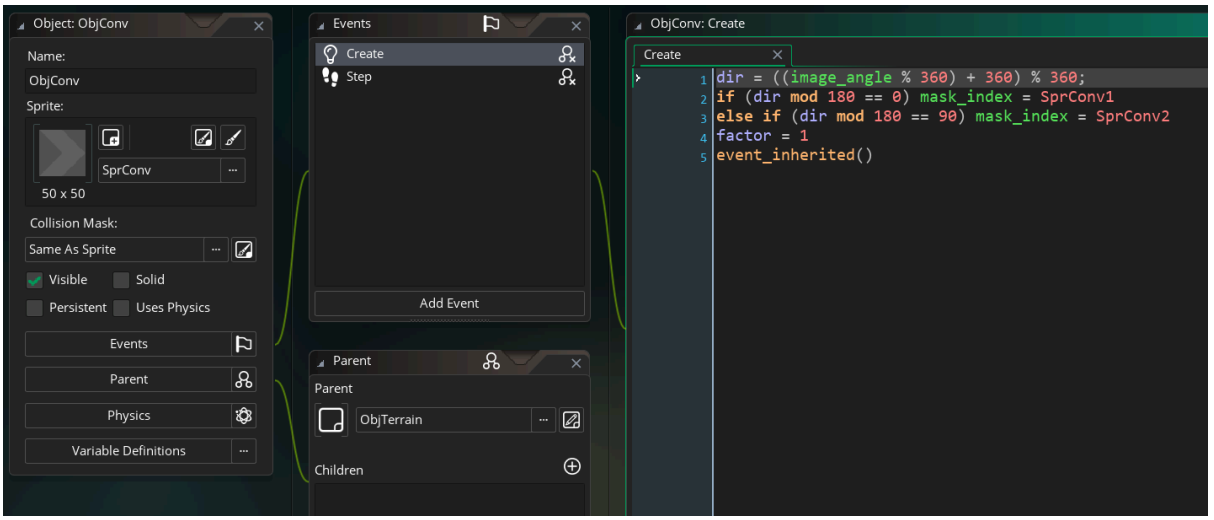

Object: ObjCheck
Name:
ObjCheck
Sprite:
SprCheck
50 x 50
Collision Mask:
Same As Sprite
Visible
Solid
Persistent
Uses Physics
Events
Parent
Physics
Variable Definitions
Events
Create
Step
Add Event
Parent
Parent
ObjTerrain
Children
Code
Create
Step
image_speed = room_speed * delta_time / 1000000
event_inherited()
INS

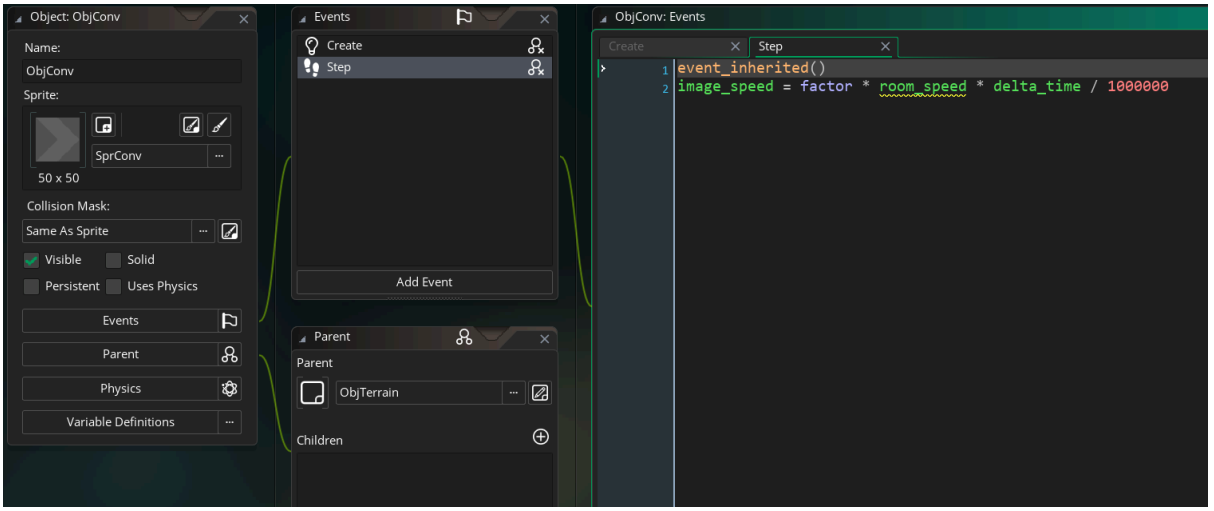

Object: ObjPit
Name:
ObjPit
Sprite:
SprPit
50 x 50
Collision Mask:
Same As Sprite
Visible
Solid
Persistent
Uses Physics
Events
Parent
Physics
Variable Definitions
Events
Create
Step
Add Event
Parent
Parent
ObjTerrain
Children
ObjTerrain: Create
Create
platform = instance_place(x, y, ObjPlatform)
if (platform != noone) {
offset_x = platform.x - x
offset_y = platform.y - y
}

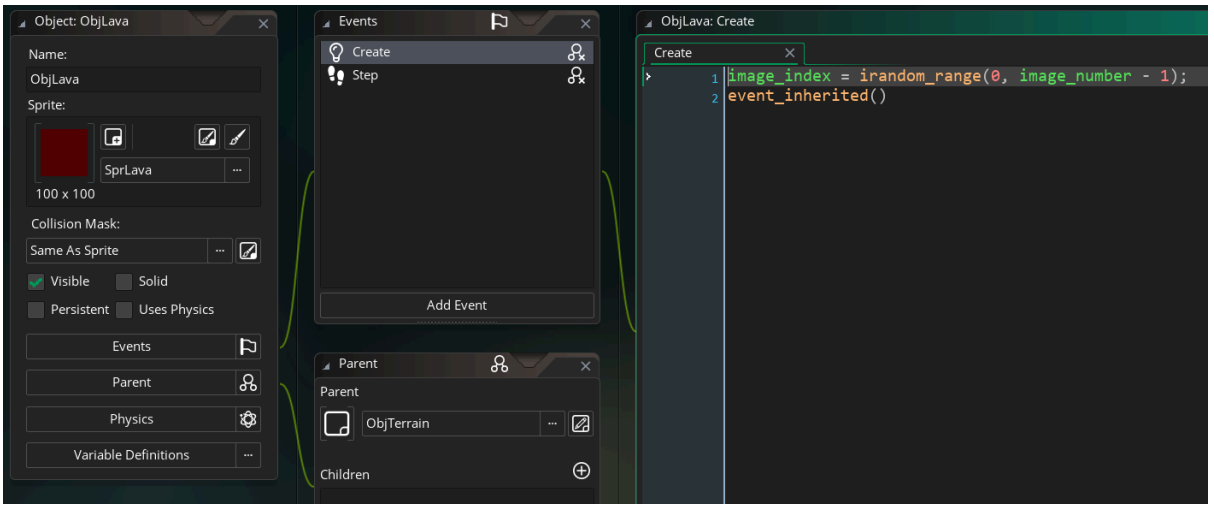

Object: ObjPit
Name:
ObjPit
Sprite:
SprPit
50 x 50
Collision Mask:
Same As Sprite
Visible
Solid
Persistent
Uses Physics
Events
Parent
Physics
Variable Definitions
Events
Create
Step
Add Event
Parent
Parent
ObjTerrain
Children
ObjTerrain: Events
Create
Step
if (platform != noone && mobile) {
x = platform.x + offset_x
y = platform.y + offset_y
}

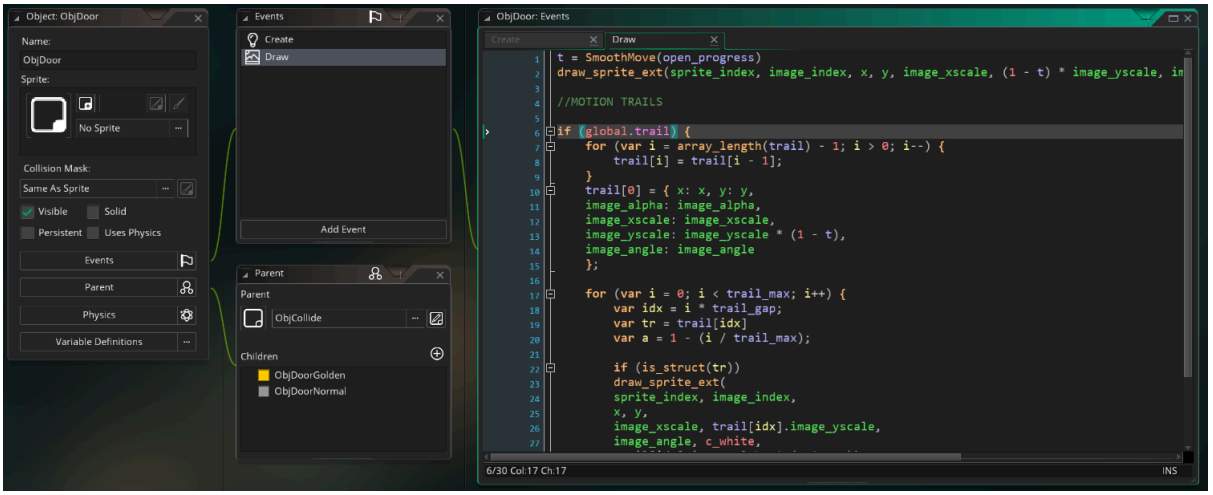

Object: ObjWater
Name:
ObjWater
Sprite:
No Sprite
Collision Mask:
Same As Sprite
Visible
Solid
Persistent
Uses Physics
Events
Parent
Physics
Variable Definitions
Events
Create
Step
Add Event
Parent
Parent
ObjTerrain
Children
ObjWaterT1
ObjWaterAll2
ObjWaterTL
ObjWaterTB
ObjWaterTRB1
ObjTerrain: Create
Create
platform = instance_place(x, y, ObjPlatform)
if (platform != noone) {
offset_x = platform.x - x
offset_y = platform.y - y
}


Object: ObjWater
Name:
ObjWater
Sprite:
No Sprite
Collision Mask:
Same As Sprite
Visible
Solid
Persistent
Uses Physics
Events
Parent
Physics
Variable Definitions
Events
Create
Step
Add Event
Parent
Parent
ObjTerrain
Children
Code
Create
Step
image_speed = room_speed * delta_time / 1000000
event_inherited()


Object: ObjConv
Name:
ObjConv
Sprite:
SprConv
50 x 50
Collision Mask:
Same As Sprite
Visible
Solid
Persistent
Uses Physics
Events
Parent
Physics
Variable Definitions
Events
Create
Step
Add Event
Parent
Parent
ObjTerrain
Children
ObjConv: Create
Create
dir = ((image_angle % 360) + 360) % 360;
if (dir mod 180 == 0) mask_index = SprConv1
else if (dir mod 180 == 90) mask_index = SprConv2
factor = 1
event_inherited()




## Ocorrência de polimorfismo

A função SmoothMove é chamada em cada um dos tipos de porta (filhos de ObjDoor: ObjDoorGolden e ObjDoorNormal)